App Final

28/04/2026

45 Points Possible

Attempt 1

NEXT UP: Review Feedback

Offline Score: N/A

Add comment

Unlimited Attempts Allowed

05/05/2026

Details

In this assessment the student **is expressly permitted** to use AI tools in the creation of content for their work **and** to proofread their work (alternatively, they can use a proofreader or non-GenAI proofreading service)

Weighting %:	**45**	**Submission deadline** (for students): as canvas
Authorship:	Individual	**Target date for returning marked coursework**: (4 weeks after submission deadline)
Tutor setting the work:	Steve Bennett	**Number of hours you are expected to work on this assignment**: 80

This Assignment assesses the following module Learning Outcomes (from Definitive Module Document):

- Demonstrate knowledge and understanding of principles of mobile operation and usability.
- Demonstrate knowledge and understanding of development and evaluation practices in mobile development.

Assignment Tasks:

You should write a mobile app in which the user is provided with a set of information, and through the app, transforms it into something else, and it is something that a credible user might find useful.

For example (this literally means *for example*, ***you do not have to*** use these examples, and because they are well trodden, they are not likely to get you the highest marks - please think of something important and specific to you).. as I was saying, for example:

Information Provided	Transformed Into	Further Enhancements
A list of meals for an online restaurant	An order	Adding to favourites
A list of football players	A fantasy team	List of results
A list of tourist attractions in city	A visit plan	Ticking off places as they are visited
A list of food in a supermarket	A shopping list	Ticking off items as they are placed in basket
A list of pokemon go characters	A list of ones you have seen	List of next targets
A list of board games	A list of ones you have seen	List of favorite games to get
A list of courses at a University	A list of ones you have selected	Adding and reading reviews
A list of restaurants in a city	Making a shortlist	Adding and reading reviews

Therefore it is an app which takes information and enriches it

For all of these things, there should be ways of navigating the information presented - by sorting, filtering, paging etc. Please feel free to email the tutors with other ideas.

Here are the rules:

You may ONLY use React Native, and the app has to function as an Expo Snack. The Snack URL must not ever be revealed to any other student and be only shared with the tutor. The initial data your app uses must be encoded in JSON and be either **a)** hard-wired into your App or **b)** come from an online source requiring no authentication.

Let me say that again: it has to be an Expo Snack. ***Not an expo project run locally***. It must be an Expo Snack. This is the only way we can manage the marking load. If it is not an Expo snack it simply wont be marked, and therefore you will score 0.

Two further requirements:

- The app MUST NOT be contained inside a single app.js file. Navigation containers, separate screens, and any custom components must have their own .js files.

- The app should not require "signup/register" process. Any logins/passwords should be created in advance and appear on the screen where they are to be used.

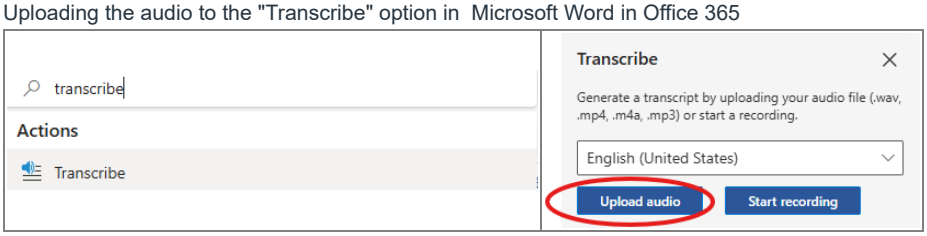
There will be an automatic 20% deduction from your score if these are not observed.

You may use web services, but since expo snacks by nature are not private, then if you are using any tokens/authentication, your credentials will be insecure. The best work around for this is to use your own webspace and potentially create your own service (not requiring authentication) which will securely poll the external service for the information it needs, and then just forward the response back. Clearly you should disable this service after the marking period is over.

For the purposes of saving data or preferences in your app, just use React's AsyncStorage. We do not require anything beyond this.

Submission Requirements (and please upload these as separate items, not all contained in one zip file):

1. The URL of the expo.dev Snack
2. The URL of any service used, which should be freely visible by tutors all up until the module board of examiners. Including your own as per the advice above.
3. A zip file of the code
4. A six minute video ***with commentary recorded in your own voice*** with the following structure
   - From 0s-60s into the video (one minute duration)- very quick run through of screens - do NOT give unnecessary background here
   - From 60s-180s into the video (two minutes duration) >= 2 comparisons of your app at an earlier stage of development to what it is at the end (These comparisons should be based on the 2 of the 3 screen capture images made for the prototype) These should explain your design decisions (including widgets used and why) and demonstrate how your thinking evolved over the project. It should also talk of alternatives you considered while making those decisions.
   - From 180s-300s into the video (two minutes duration) A detailed analysis of a snippet of code in your app, describing the data/data structures it contains, the operations performed up on it, and how it is rendered on screen
   - From 300s-360s (one minute duration) - Specify the main thing you learned while doing this work
5. A transcript of your narrated video. This is easily obtained by
   1. Extracting the audio from the video (see **How can I extract audio from video with ffmpeg? - Stack Overflow (https://stackoverflow.com/questions/9913032/how-can-i-extract-audio-from-video-with-ffmpeg)**)
   2. Uploading the audio to the "Transcribe" option in Microsoft Word in Office 365



These time specifications can vary a little so dont worry about any section being 15% under or above the recommended duration for that section. However, the markers are not obliged to listen beyond the 360s though they may do so if your video is engaging enough. If it is not, they will just stop watching - and no appeal for marks unawarded because of overlong duration of video will be entertained.

You should be careful not to share code (and specifically snack urls) with other students on the course. If you fear you have shared one, save it to a new name and (after having checked that the new named version is correct, delete the original.

*Please make sure your video is below 100mb

Marks awarded for:

Mobile Computing Final App Rubric (2025)

Criteria	Ratings	
Artefact: Functionality and Usability:	15 to >12.0 Excellence-> V.Good an brilliant app, superb both at the level of the idea itself and also its execution in terms of design and usability. 12 to >9.0 Excellence-> V.Good a really good app, an interesting idea which is carried out very well - there may be some minor niggles at the level of execution 9 to >6.0 Good->BarePass an app with some really good features, but logic or usability or design issues, or a very good, but very generic, app 6 to >3.0 BarePass ->Poorrather simple, but does demonstrate fundamentals of mobile computing, also for extremely generic work 3 to >0 Poor->V.Poor too simple, too incomplete, or too buggy	15
Video:App Comparison	15 to >12.0 Excellence-> V.Good multiple comparisons, between earlier and later state, comprising the challenge of execution, involving real reflection which might digress on to theories of usability or development paradigms 12 to >9.0 V.Good ->Good a real demonstration of the app at an earlier stage, and the current stage, the reason for the difference, and the challenges of execution - and detailed, convincing and uncommon. 9 to >6.0 Good->BarePass a real demonstration of the app at an earlier stage, and the current stage, the reason for the difference, and the challenges of execution. This however might be slightly obvious 6 to >3.0 BarePass ->Poor a banal story of items missing and items later added (previous it didn't have x, now it has x) 3 to >0 Poor->V.Poor no real comparison made, or one which is extremely cursory	15
Video: Personal	15 to >12.0 Excellence-> V.Good a brilliant narration of a development journey, which expands beyond the specifics of this course, to encompass other areas of computer science	15

Criteria	Ratings	
Development Journey	12 to >9.0 V.Good ->Good a really good isolation of moments of difficulty highly specific to the app and its field and goals, with a description of solutions attempted and growth of understanding arising therefrom 9 to >6.0 Good->BarePass a narration of the development process which connects moments of blockage during development, solutions obtained, and greater understanding resulting therefrom. These however might be slightly predicatable. 6 to >3.0 BarePass ->Poor a narration of things done, but the description growing understanding either lacks credibility or speaks in very abstract terms without demonstrating in specific detail with reference to the app 3 to >0 Poor->V.Poor lacking personality, just a narration of things done	
Video: Architecture and Tradeoffs	15 to >12.0 Excellence-> V.Good describes the architecture of the app at the appropriate levels (datastructures and business logic), and describes alternative possibilities. Describes the solution chosen, and evaluates how that worked out (you will not be penalized for saying you took the wrong path). Levels appropriate, alternatives credible, evaluation meaningful. 12 to >9.0 V.Good ->Good describes well the architecture of the app and thinks of alternatives in detailed ways. Establishes pros and cons for both approaches. Describes the solution chosen, and evaluates how that worked out. Maybe not so strong in one of the elements (levels, alternatives, evaluation) 9 to >6.0 Good->BarePass describes architectural tradeoffs but might be for problems of a lesser order, or the consequences of either solution being chosen might not be particularly important, or couched at the wrong level 6 to >3.0 BarePass ->Poordescribes a tradeoff at too high a level (in architectural terms, or describes something having no real consequence, or maybe concentrate on very trivial things (say the iconography) which is not at an architectural level 3 to >0 Poor->V.Poor Doesnt really communicate the sense that the app could have been written in any other way than the way it was	15
Submission Completeness	5 to >0.0 Full marks 0 No marks	5

Type of Feedback to be given for this assignment: **Written**

Additional information:

- Regulations governing assessment offences including Plagiarism and Collusion are available from **https://www.herts.ac.uk/__data/assets/pdf_file/0007/237625/AS14-Apx3-Academic-Misconduct.pdf (https://www.herts.ac.uk/__data/assets/pdf_file/0007/237625/AS14-Apx3-Academic-Misconduct.pdf)** (UPR AS14).
- Guidance on avoiding plagiarism can be found here: **https://herts.instructure.com/courses/61421 (https://herts.instructure.com/courses/61421)** (see the Referencing section)
- For undergraduate modules:
- a score of 40% or above represents a pass performance at honours level.
- late submission of any item of coursework for each day or part thereof (or for hard copy submission only, working day or part thereof) for up to five days after the published deadline, coursework relating to modules at Levels 0, 4, 5, 6 submitted late (including deferred coursework, but with the exception of referred coursework), will have the numeric grade reduced by 10 grade points until or unless the numeric grade reaches or is 40. Where the numeric grade awarded for the assessment is less than 40, no lateness penalty will be applied.
- For postgraduate modules:
- a score of 50% or above represents a pass mark.
- late submission of any item of coursework for each day or part thereof (or for hard copy submission only, working day or part thereof) for up to five days after the published deadline, coursework relating to modules at Level 7 submitted late (including deferred coursework, but with the exception of referred coursework), will have the numeric grade reduced by 10 grade points until or unless the numeric grade reaches or is 50. Where the numeric grade awarded for the assessment is less than 50, no lateness penalty will be applied.

View Rubric

Mobile Computing Final App Rubric (2025)

Criteria	Ratings					Points
Artefact: Functionality and Usability:	**10 to >8 pts Excellence->V.Good** an brilliant app, superb both at the level of the idea itself and also its execution in terms of design and usability.	**8 to >6 pts Excellence->V.Good** a really good app, an interesting idea which is carried out very well - there may be some minor niggles at the level of execution	**6 to >4 pts Good->BarePass** an app with some really good features, but logic or usability or design issues, or a very good, but very generic, app	**4 to >2 pts BarePass->Poor** rather simple, but does demonstrate fundamentals of mobile computing, also for extremely generic work	**2 to >0 pts Poor->V.Poor** too simple, too incomplete, or too buggy	/ 10 pts
Video:App Comparison	**10 to >8 pts Excellence->V.Good** multiple comparisons, between earlier and later state,	**8 to >6 pts V.Good->Good** a real demonstration of the app at an earlier stage, and the current stage, the	**6 to >4 pts Good->BarePass** a real demonstration of the app at an earlier stage, and the current stage, the	**4 to >2 pts BarePass->Poor** a banal story of items missing and items later added (previous it didn't	**2 to >0 pts Poor->V.Poor** no real comparison made, or one which is extremely cursory	/ 10 pts

Mobile Computing Final App Rubric (2025)

Criteria	Ratings					Points
	comprising the challenge of execution, involving real reflection which might digress on to theories of usability or development paradigms	reason for the difference, and the challenges of execution - and detailed, convincing and uncommon.	reason for the difference, and the challenges of execution. This however might be slightly obvious	have x, now it has x)		
Video: Personal Development Journey	**10 to >8 pts Excellence->V.Good** a brilliant narration of a development journey, which expands beyond the specifics of this course, to encompass other areas of computer science	**8 to >6 pts V.Good->Good** a really good isolation of moments of difficulty highly specific to the app and its field and goals, with a description of solutions attempted and growth of understanding arising therefrom	**6 to >4 pts Good->BarePass** a narration of the development process which connects moments of blockage during development, solutions obtained, and greater understanding resulting therefrom. These however might be slightly predicatable.	**4 to >2 pts BarePass->Poor** a narration of things done, but the description growing understanding either lacks credibility or speaks in very abstract terms without demonstrating in specific detail with reference to the app	**2 to >0 pts Poor->V.Poor** lacking personality, just a narration of things done	/ 10 pts
Video: Architecture and Tradeoffs	**10 to >8 pts Excellence->V.Good** describes the architecture of the app at the appropriate levels (datastructures and business logic), and describes alternative possibilities. Describes the solution chosen, and evaluates how that worked out (you will not be penalized for saying you took the wrong path). Levels appropriate, alternatives credible, evaluation meaningful.	**8 to >6 pts V.Good->Good** describes well the architecture of the app and thinks of alternatives in detailed ways. Establishes pros and cons for both approaches. Describes the solution chosen, and evaluates how that worked out. Maybe not so strong in one of the elements (levels, alternatives, evaluation)	**6 to >4 pts Good->BarePass** describes architectural tradeoffs but might be for problems of a lesser order, or the consequences of either solution being chosen might not be particularly important, or couched at the wrong level	**4 to >2 pts BarePass->Poor** describes a tradeoff at too high a level (in architectural terms, or describes something having no real consequence, or maybe concentrate on very trivial things (say the iconography) which is not at an architectural level	**2 to >0 pts Poor->V.Poor** Doesnt really communicate the sense that the app could have been written in any other way than the way it was	/ 10 pts
Submission Completeness	**5 to >0 pts Full marks**		**0 pts No marks**			/ 5 pts
						Total points: 0

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